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Growth is measurable, it is amazing to know that one single maize root apical meristem can give rise to more than 17,500 new cells per hour and cells in a watermelon may increase in size upto 3,50,000 times.

- The primary growth of the plant is due to the activity of apical meristem where, new cells are added to root and shoot apex causing linear growth of plant body.
- The secondary vascular cambium and cork cambium add new cells to cause increase in girth.
- Leaves, flowers and fruits are limited in growth or of determinate or **closed form growth.**
- Monocarpic annual plants produce flowers only once during lifetime and dies. Example: Paddy and Bean
- Monocarpic perennials produce flowers only once during life time but the plants survive for many years. Example: Bamboo.
- Polycarpic perennials produce flowers every year during life time. Example: Coconut.

15.1.1 Indication of growth

Growth in plants can be measured in terms of,

i. Increase in length or girth (roots and stems)
ii. Increase in fresh or dry weight
iii. Increase in area or volume (fruits and leaves)
iv. Increase in number of cells produced.

15.1.2 Phases of growth

There are three phases of growth,

1. Formative phase
2. Elongation phase
3. Maturation phase

1. Formative phase: Growth in this phase occurs in meristematic cells of shoot and root tips. These cells are small in size, have dense protoplasm, large nucleus and small vacuoles. Cells divide continuously by mitotic cell division. Some cells retain capability of cell division while other cells enter the next phase of growth (Figure 15.1).

2. Elongation Phase: Newly formed daughter cells are pushed out of the meristematic zone and increases the volume. It requires auxin and food supply, deposition of new cell wall materials (intussusception), addition of protoplasm and development of central vacuole take place.

3. Maturation Phase: During this stage cells attain mature form and size. Thickening and differentiation takes place. After differentiation, the cells do not grow further.

Activity

Demonstration of phases of growth

To demonstrate and study the phases of growth, germinate a few seeds of bean on a circular filter paper soaked with water in a petridish. After two days of growth, select a few seedlings with straight radical of 2 to 3 cm length. Dry the surface of radical with a blotting paper and mark the radical from tip to base with at least 2 mm gap using water proof ink. Replace the seedlings in filter paper and observe further growth.